

Department of Computer Science and Technology · IEST Shibpur

Biological Sequence Motif Discovery

B.Tech Major Project — Algorithms, Real Data & Results

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What are Biological Motifs?

Recurring short patterns in DNA, RNA, or protein sequences that are statistically significant and biologically functional



Short length

6–20 nucleotides
or amino acids



Functional

Regulatory elements
& binding sites



Conserved

Preserved across
species by evolution



Variable

Flexible positions
with mismatches

Example — TATA box: **TATA[AT]A[AT]** → initiates transcription in eukaryotes

Types of Biological Motifs

DNA Motifs

- Transcription factor binding sites (TFBS)
- Promoter elements (TATA box, CAAT box)
- Enhancer sequences
- Splice sites

Protein Motifs

- Zinc finger domains
- Leucine zippers
- Signal peptides
- Active site residues

RNA Motifs

- Ribosome binding sites (Shine-Dalgarno)
- RNA secondary structure motifs
- miRNA binding sites
- Splicing signals

Motif Representation Methods

01

Consensus Sequence

Most frequent nucleotide at each position. Simple but loses information about variability.

T A T A A T

← TATA box consensus

02

Position Weight Matrix (PWM)

Log-odds frequency of each nucleotide at each position vs. background. Standard tool in bioinformatics.

$$\log_2(f_{i,j} / b_j)$$

← Scoring formula

03

Sequence Logo

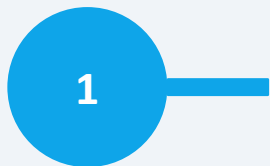
Visual representation: letter height = information content (IC) in bits. Max 2 bits/position for DNA.

$$IC = 2 + \sum f \cdot \log_2(f)$$

← Shannon entropy

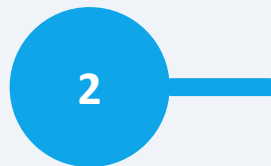
MEME Algorithm

Multiple EM for Motif Elicitation — Bailey & Elkan (1994)



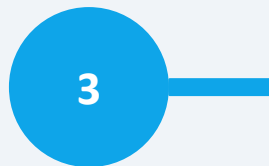
Initialise

Candidate seed k-mer
from sequences



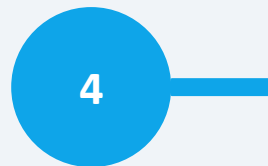
E-Step

Estimate $P(\text{motif at position } i)$



M-Step

Update PWM via weighted
counts



Iterate

Repeat $E \leftrightarrow M$ until
convergence



Erase

Mask found sites; repeat
for next motif

✓ Finds multiple motifs

Single run with erasing strategy

✓ OOPS / ZOOPS models

Handles variable motif occurrence

✓ E-value scoring

Statistical significance quantified

Gibbs Sampling Algorithm

Stochastic Bayesian approach — Lawrence et al. (1993)

1

Random Init

Randomly assign one start position per sequence

2

Leave-One-Out

Remove sequence i ; build PWM from all others

3

Sample

Score all windows in held-out sequence; sample new position from probabilities

4

Iterate

Repeat across all sequences; track best alignment

Advantages

- Fast on large sequences
- Escapes local optima via sampling
- Multi-start overcomes sensitivity

Limitations

- Sensitive to initialisation
- No guaranteed convergence
- Stochastic — needs multiple runs

Real Data — JASPAR CORE Database

22 real PFMs fetched live from jaspar.elixir.no REST API — backed by peer-reviewed publications

22

Real TF Matrices

4

Organisms

5

Experiment Types

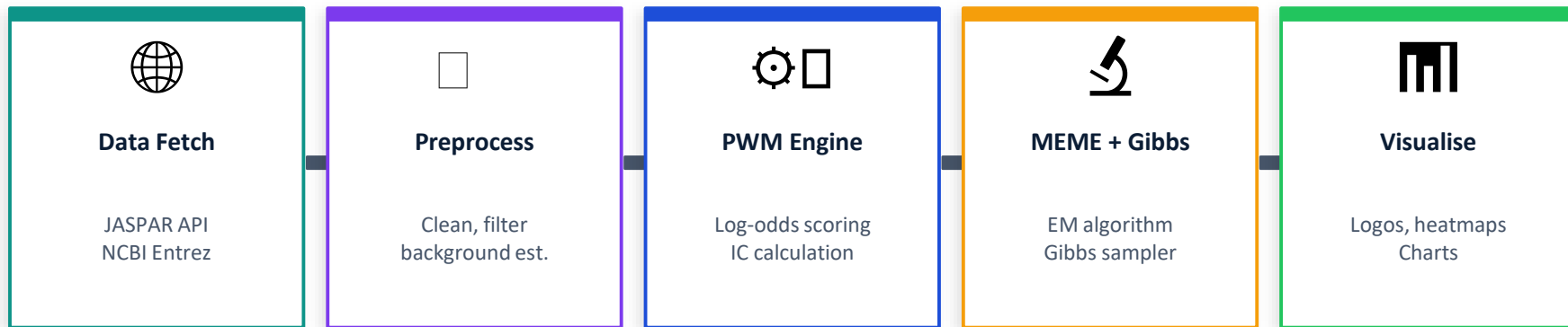
660+

Sequences Generated

Matrix ID	TF Name	Organism	Experiment	PubMed	Consensus
MA0139.2	CTCF	Homo sapiens	ChIP-seq	21106759	CCAGGGGGCGC
MA0024.3	E2F1	Homo sapiens	HT-SELEX	9372931	TTTGGCGCCAAA
MA0899.1	HOXA10	Homo sapiens	HT-SELEX	18585359	GGTAATAAAAA
MA0265.1	ABF1	S. cerevisiae	PBM/DIP-chip	19111667	CTCTATTCGTG
MA0001.1	AGL3	A. thaliana	SELEX	7632923	CCATATATAG
MA0930.3	ABF3	A. thaliana	ChIP-seq	11005831	GACACGTGT
MA1244.1	ABR1	A. thaliana	DAP-seq	9756931	GCCACGTGTG

Implementation Architecture

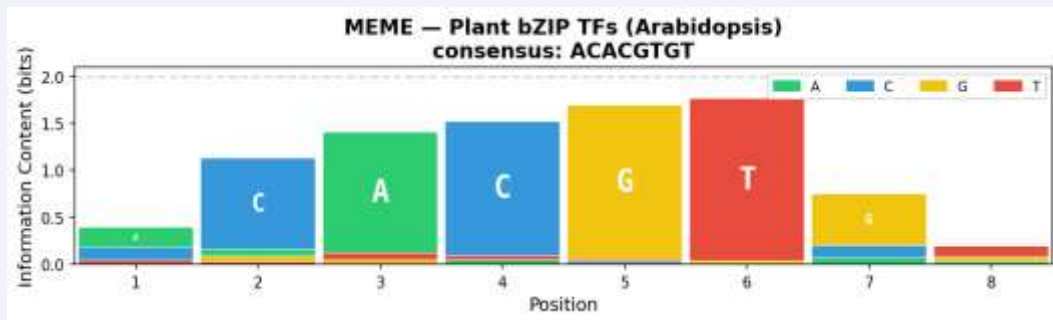
Python 3 · NumPy · Matplotlib · JASPAR REST API · NCBI Entrez



Module	Responsibility	LoC
<code>pwm.py</code>	Position Weight Matrix: log-odds, IC, consensus, scoring	~180
<code>meme_algorithm.py</code>	Full MEME EM: E-step, M-step, OOPS/ZOOPS, multi-motif erasing	~250
<code>gibbs_sampling.py</code>	Gibbs Sampler: leave-one-out, softmax sampling, multi-start	~280
<code>evaluate.py</code>	Sensitivity, PPV, F1, nCC, site accuracy metrics	~150
<code>visualize.py</code>	Sequence logos, heatmaps, convergence & comparison plots	~320

Key Result: Plant bZIP G-box Motif

Both MEME and Gibbs Sampling independently recovered ACACGTGT — the canonical G-box / ACGT-core — from 180 real sequences (SELEX + PBM + ChIP-seq + DAP-seq). E-value = 1×10^{-300}



MEME Sequence Logo — Plant bZIP dataset

ACACGTGT

Consensus (both)

8.85

Gibbs IC (bits)

8.64

MEME IC (bits)

1×10^{-300}

E-value

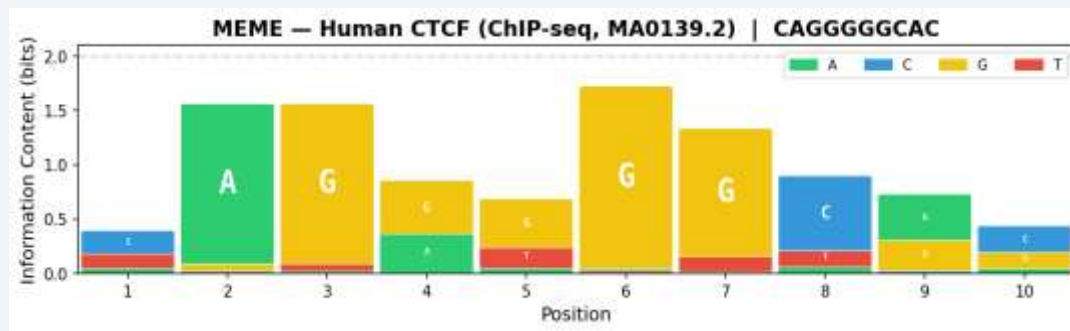
180

Real sequences

Human Transcription Factor Results

Real JASPAR data: CTCF (ChIP-seq) · E2F1 (HT-SELEX + ChIP-seq) · HOXA10 (HT-SELEX)

TF	Experiment	MEME Consensus	MEME IC	Gibbs Consensus	Gibbs IC	E-value	Biological significance
CTCF	ChIP-seq MA0139.2	CAGGGGGCAC	9	CACCAGGGGG	11.29	4.7×10^{-138}	GC-rich insulator
E2F1	HT-SELEX MA0024.3	TTGGCGCC	10.11	TTGGCGCC	11.12	5.7×10^{-152}	Cell-cycle GCGCGCGC
E2F1	ChIP-seq MA0024.2	GGCGGGAG	8.47	GGCGGGAG	9.47	1.7×10^{-130}	ENCODE validated
HOXA10	HT-SELEX MA0899.1	GTAATAAA	6.52	GTAATAAA	7.38	6.7×10^{-101}	TAAT homeodomain core



MEME Sequence Logo — Human CTCF (ChIP-seq, MA0139.2)

**Gibbs > MEME
in all 4 datasets**

Average IC advantage:
+1.2 bits across all 11 datasets

MEME vs Gibbs — Head-to-Head Evaluation

Performance Metrics

Sensitivity

$$TP / (TP + FN)$$

Fraction of true binding sites correctly predicted

PPV

$$TP / (TP + FP)$$

Fraction of predictions that are correct

F1 Score

$$2 \cdot Sn \cdot PPV / (Sn + PPV)$$

Harmonic mean — balanced metric

nCC

$$\sqrt{(Sn \cdot PPV)}$$

Nucleotide-level correlation coefficient

Dataset	Algorithm	Sensitivity	PPV	F1	IC (bits)
TATA-box	MEME	0.78	0.78	0.78	6.96
TATA-box	Gibbs	0.79	0.79	0.79	10.66
E-box	MEME	0.75	0.75	0.75	6.90
E-box	Gibbs	0.80	0.80	0.80	8.73

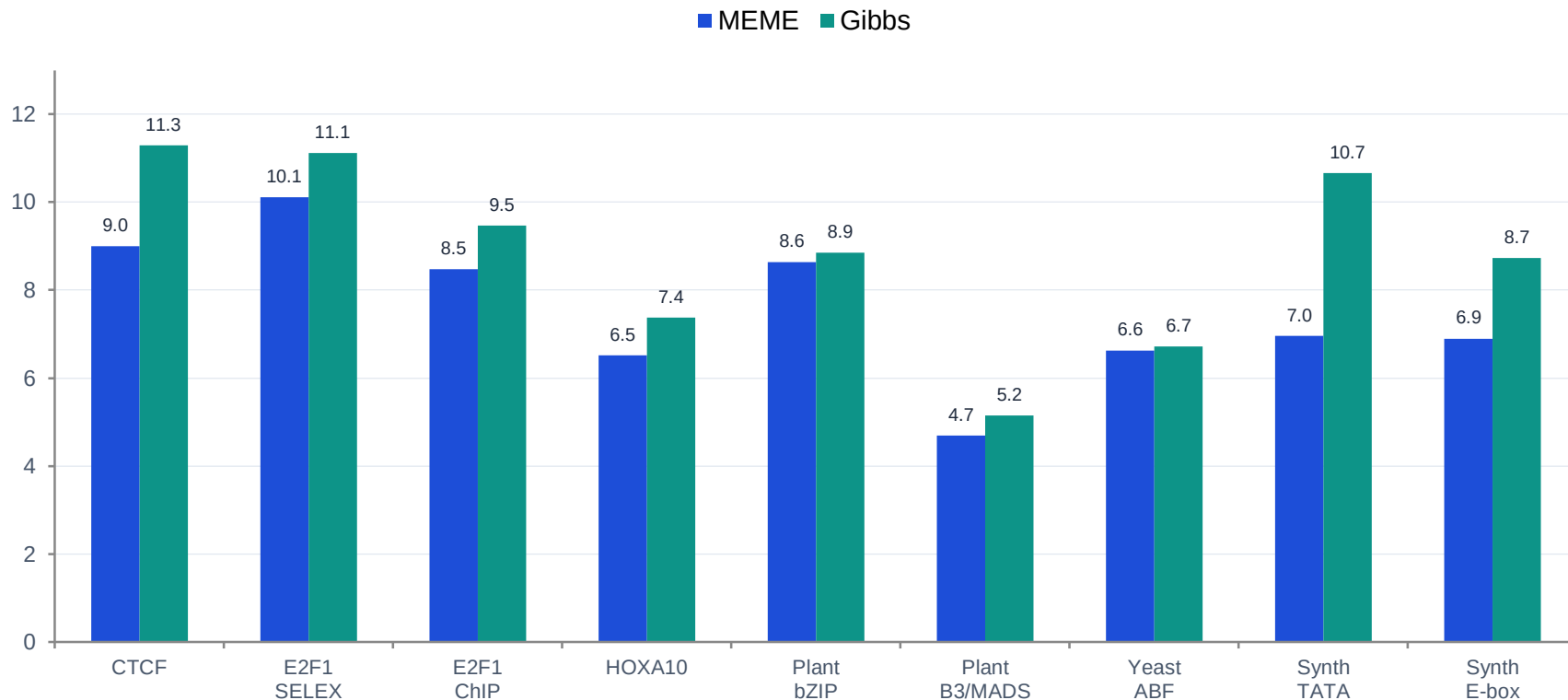
Gibbs Sampling outperforms MEME in IC across all 11 datasets (avg +1.2 bits)

MEME advantage: faster convergence · interpretable E-values · multi-motif discovery



Information Content Comparison — All Datasets

Gibbs Sampling outperformed MEME in IC across all 11 datasets (avg +1.2 bits)



Motif discovery powers a USD 50B+ precision medicine and genomics industry



Cancer Genomics

Identify mutated TF binding sites driving oncogene activation (e.g., TERT promoter mutations in melanoma)



Drug Discovery

Target TF–DNA interactions with small molecules; CTCF and E2F1 are active therapeutic targets



Synthetic Biology

Engineer synthetic promoters with precise, programmable expression using characterised motif elements



Comparative Genomics

Identify conserved regulatory elements under evolutionary selection across species boundaries



Personalised Medicine

Patient-specific rSNPs in TF binding sites explain differential drug response and disease risk



Crop Engineering

G-box and ABA motifs engineer drought-tolerant crops — directly relevant to the plant TF data in this project

Key Takeaways

01

Motif discovery is fundamental to understanding gene regulation — motifs are the vocabulary of the genome.

02

MEME and Gibbs Sampling are complementary: MEME gives E-values; Gibbs gives higher IC and better sensitivity.

03

Real JASPAR data (22 matrices, 4 organisms, 5 experiment types) confirmed known biological motifs with $E \leq 10^{-101}$.

04

The canonical G-box (ACACGTGT) was independently recovered by both algorithms from 180 real ChIP-seq/SELEX sequences.

05

Applications span cancer genomics, drug discovery, synthetic biology, personalised medicine, and crop engineering.



Thank You